

1. Definition and Meaning of Money

Money is anything **generally accepted in payment for goods, services, or debts.**

- **Currency** = coins and bills.
 - **Wealth** = stock of assets (money, bonds, land, houses, etc.).
 - **Income** = flow of money or payments over time.
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2. Theories of the Value of Money

a. Quantity Theory

- **Velocity (V)**: the rate at which money circulates.

Formula:

$$v = \frac{T}{M}$$

or using GDP:

$$M \times V = P \times Y$$

where **M** = money supply, **P** = price level, **Y** = output.

- When **V** is constant, money supply determines **nominal GDP (P×Y)**.
- **Real GDP** depends on resources (capital & labor), while **price level** adjusts accordingly.

b. Cambridge Theory

Connects **money demand** with **income**:

$$\left(\frac{M}{P^4}\right) = K \times Y$$

where **k** = 1/V, meaning people hold more money (high k) when money circulates slowly (low V).

c. Friedman and Keynesian Views

- Friedman emphasized long-term relation between **money supply** and **inflation**.
- Keynesian equations explain how changes in money affect income, interest, and employment.

3. Connection Between Money and Prices

- **Inflation rate** = % increase in average price level.
- **Price** = amount of money required to buy goods.
- As prices are defined in money terms, understanding **money's nature and control** is crucial to understanding inflation.

4. Functions of Money

1. **Medium of Exchange** – simplifies trade, replaces barter.
2. **Unit of Account** – measures and compares value of goods.
3. **Store of Value** – maintains purchasing power over time.
 - People choose money for **liquidity**, not for return.

5. Liquidity

Liquidity = how easily an asset can be converted into cash.

- **Advantages:** low transaction cost, fast access.
- **Disadvantages:** no interest and inflation reduces its value.

6. Measuring Money (Monetary Aggregates)

Used by **central banks** to track money supply.

- **M1:** cash and assets used directly as medium of exchange.
 - **M2:** M1 + very liquid short-term assets.
 - **M3:** M2 + less liquid, long-term assets.
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7. Financial Markets and Their Function

Financial markets **channel funds**:

- **Direct finance**: borrowers sell securities directly to lenders.
- **Indirect finance**: via **financial intermediaries** like banks.

Importance:

- Gives savers interest income.
 - Helps investors earn profits and promote growth.
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8. Debt and Equity Markets

- **Debt instruments (bonds)**: pay interest until maturity.
- **Equity (stocks)**: represent ownership and share in profit.

Maturity Types:

- Short-term: <1 year
- Intermediate: 1–10 years
- Long-term: >10 years

Equities are **only long-term**.

9. Primary vs. Secondary Markets

- **Primary market**: where new securities are issued (firms raise funds).
 - **Secondary market**: where existing securities are traded (no new funds raised).
 - Organized as **Exchanges** or **Over-the-counter markets (OTC)**.
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10. Money vs. Capital Markets

- **Money Market**: short-term securities (<1 year).
- **Capital Market**: long-term securities (>1 year, e.g., bonds, stocks).

11. Financial Market Instruments

Money Market Instruments:

- **Treasury Bills** – short-term government debt, sold at discount.
- **Certificates of Deposit (CDs)** – bank debt paying interest.
- **Commercial Paper** – issued by big corporations.
- **Repurchase Agreements (REPOs)** – short-term loans backed by securities.
- **Federal Funds** – overnight loans between banks.

Capital Market Instruments:

- **Corporate Stocks** – ownership claims in firms.
- **Mortgages** – loans for property, secured by collateral.
- **Mortgage-Backed Securities** – pooled mortgages sold to investors.
- **Corporate Bonds** – long-term bonds by firms.
- **Convertible Bonds** – can turn into stock.
- **Government Securities** – long-term debt issued by the state.

12. Financial Intermediaries

Three main types:

1. **Depository institutions:** commercial banks, savings & loans, credit unions.
2. **Contractual saving institutions:** pension funds, insurance companies.
3. **Investment intermediaries:** finance companies, mutual funds, money market funds.

13. Key Macroeconomic Concepts

- **GDP:** total value of goods and services produced.
- **Aggregate income:** total earnings of factors (land, labor, capital).
- **Nominal vs. Real:** nominal uses current prices, real uses constant prices.
- **GDP Deflator:** $\frac{\text{Nominal GDP}}{\text{Real GDP}}$
- **Growth Rate:** measures % change in GDP.
- **Inflation Rate:** measures % change in price level.